chinmaya@chinmaya-HP-EliteBook-x360-1040-G7-Notebook-PC: ~/OS_Project/G4_Project1_xv6CustomizeSystemCalls/xv6-riscv

GNU nano 7.2 kernel/sysproc.c

```
// ---------------------------------------------------
// Sujitha's Work
// ---------------------------------------------------
uint64 sys_mutex_lock(void){
    mutex_acquire(&m);
    return 0;
}

uint64 sys_mutex_unlock(void){
    mutex_release(&m);
    return 0;
}
uint64 sys_mutex_test_inc(void){
    mutex_acquire(&m);
    counter++;
    mutex_release(&m);
    return 0;
}

uint64 sys_get_counter(void){
    return counter;
}


// -------------------------------------------------------
// Chinmaya's Work
// -------------------------------------------------------
// For threads
uint64 sys_clone(void){
    return sys_fork();   // use existing stable syscall
}

uint64 sys_join(void){
    return sys_wait();  // use existing stable syscall
}

// -------------------------------------------------------
```

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^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line M-E Redo M-6 Copy ^Q Where Was M-W Next ^F Forward

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GNU nano 7.2 user/testthread.c

```
#include "kernel/types.h"
#include "user/user.h"

void thread_func(){
        printf("Hello from thread!\n");
        exit(0);
}

int main(){
        void *stack = malloc(4096);
        int pid = clone(stack);

        if(pid == 0) thread_func();
        else{
                join(&stack);
                printf("Thread finished\n");
        }

        exit(0);
}
```

[Read 20 lines]

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GNU nano 7.2 kernel/defs.h *

```
void            kvminit(void);
void            kvminithart(void);
void            kvmmap(pagetable_t, uint64, uint64, uint64, int);
int             mappages(pagetable_t, uint64, uint64, uint64, int);
pagetable_t     uvmcreate(void);
uint64          uvmalloc(pagetable_t, uint64, uint64, int);
uint64          uvmdealloc(pagetable_t, uint64, uint64);
int             uvmcopy(pagetable_t, pagetable_t, uint64);
void            uvmfree(pagetable_t, uint64);
void            uvmunmap(pagetable_t, uint64, uint64, int);
void            uvmclear(pagetable_t, uint64);
pte_t *         walk(pagetable_t, uint64, int);
uint64          walkaddr(pagetable_t, uint64);
int             copyout(pagetable_t, uint64, char *, uint64);
int             copyin(pagetable_t, char *, uint64, uint64);
int             copyinstr(pagetable_t, char *, uint64, uint64);
int             ismapped(pagetable_t, uint64);
uint64          vmfault(pagetable_t, uint64, int);

// plic.c
void            plicinit(void);
void            plicinithart(void);
int             plic_claim(void);
void            plic_complete(int);

// virtio_disk.c
void            virtio_disk_init(void);
void            virtio_disk_rw(struct buf *, int);
void            virtio_disk_intr(void);

// number of elements in fixed-size array
#define NELEM(x) (sizeof(x)/sizeof((x)[0]))


// Chinmaya's Work
int fork(void);
int wait(uint64);
```

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^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line M-E Redo M-6 Copy ^Q Where Was M-W Next ^F Forward

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```
it -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie
 -no-pie   -c -o user/year_test.o user/year_test.c
riscv64-linux-gnu-ld -z max-page-size=4096 -T user/user.ld -o user/_year_test user/year_test.o user/ulib.o user/usys.o user/printf.o user/umalloc.o
riscv64-linux-gnu-objdump -S user/_year_test > user/year_test.asm
riscv64-linux-gnu-objdump -t user/_year_test | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > user/year_test.sym
riscv64-linux-gnu-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -march=rv64gc -MD -mcmodel=medany -ffreestanding -fno-common -nostdlib -fno-built
in-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-ex
it -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie
 -no-pie   -c -o user/msgtest.o user/msgtest.c
riscv64-linux-gnu-ld -z max-page-size=4096 -T user/user.ld -o user/_msgtest user/msgtest.o user/ulib.o user/usys.o user/printf.o user/umalloc.o
riscv64-linux-gnu-objdump -S user/_msgtest > user/msgtest.asm
riscv64-linux-gnu-objdump -t user/_msgtest | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > user/msgtest.sym
riscv64-linux-gnu-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -march=rv64gc -MD -mcmodel=medany -ffreestanding -fno-common -nostdlib -fno-built
in-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-ex
it -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie
 -no-pie   -c -o user/shmtest.o user/shmtest.c
riscv64-linux-gnu-ld -z max-page-size=4096 -T user/user.ld -o user/_shmtest user/shmtest.o user/ulib.o user/usys.o user/printf.o user/umalloc.o
riscv64-linux-gnu-objdump -S user/_shmtest > user/shmtest.asm
riscv64-linux-gnu-objdump -t user/_shmtest | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > user/shmtest.sym
riscv64-linux-gnu-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -march=rv64gc -MD -mcmodel=medany -ffreestanding -fno-common -nostdlib -fno-built
in-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-ex
it -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie
 -no-pie   -c -o user/testthread.o user/testthread.c
riscv64-linux-gnu-ld -z max-page-size=4096 -T user/user.ld -o user/_testthread user/testthread.o user/ulib.o user/usys.o user/printf.o user/umalloc.o
riscv64-linux-gnu-objdump -S user/_testthread > user/testthread.asm
riscv64-linux-gnu-objdump -t user/_testthread | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > user/testthread.sym
mkfs/mkfs fs.img README user/_cat user/_echo user/_forktest user/_grep user/_init user/_kill user/_ln user/_ls user/_mkdir user/_rm user/_sh user/_stressfs user/_usertests user/_grind
user/_wc user/_zombie user/_logstress user/_forphan user/_dorphan user/_testhello user/_testproc user/_mutextest user/_year_test user/_msgtest user/_shmtest user/_testthread
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000
balloc: first 1173 blocks have been allocated
balloc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=none,format=raw,id=x0 -device
virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 1 starting
hart 2 starting
init: starting sh
$ testthread
Hello from thread!
Thread finished
$
```